Team Weekly Sync: Project Checklist

Group: Team 1, Wellness Grocery Inventory Analysis	**Date:** October 29, 2026
Members: Jack Kelly, Anirudh Ramesh, Ikumi Uemura	**Sprint:** Week 2 (Data Audit & EDA)

Note: our team builds the pipeline in Python (allowed by the project guidelines), so KNIME items are answered with their Python equivalents.

1. Infrastructure & Accountability

	Item	Status / notes
☑	**GitHub Sync:** Did everyone pull the latest changes from main?	Yes, all three members pulled the latest version of main.
☑	**Contribution Log:** Has the README.md been updated to reflect progress made this week?	Yes, the README now documents the repo layout, the Week 2 script, and how to run it.
☑	**Blocking Issues:** Is anyone currently "blocked" by a node error or a data issue?	No, nobody is blocked. The data has no missing values, and the code runs without errors.

2. Current Sprint Focus

	Week	Status / notes
☑	**Week 2 (EDA):** Are your distributions analyzed?	**Complete.** Histograms and box plots for all 8 nutrients, Pearson and Spearman correlation matrices, a data-quality audit, and an interactive dashboard. Report: *reports/Week2_Data_Audit_EDA.pdf*.
☐	**Week 3 (Prep):** Is your feature matrix complete?	Next sprint (see action items). Week 2 already identified the correlation groups to use for feature selection.
☐	**Week 4 (Modeling):** Is the pipeline running end-to-end?	Upcoming (PCA and k-Means).
☐	**Week 5:** Validation / corrective strategies	Upcoming (silhouette scores, cluster purity, confusion matrix).
☐	**Week 6 (Presentation):** Does the workflow run end-to-end on new data?	Upcoming.

3. Technical Quality Assurance (the "Green-to-Gold" audit)

	Item	Status / notes
☑	**Automation:** Does the workflow run from reading the file to output without changing settings?	Yes. *code/week2_data_audit.py* reads *data/food.csv* and produces every figure, table, and dashboard in one run, with no manual steps.
☑	**Annotation:** Are the primary logic steps explained?	Yes, for Week 2: each step (audit, histograms, box plots, correlations, dashboard) is a commented section of the script, and the report explains each result. Normalization starts in Week 4 and will be annotated then.
☑	**Evidence:** Is every chart/table in the report linked to a specific output?	Yes. Every figure and table in the report is generated by the script and saved in *outputs/week2/*.

Week 2 key findings (for the record)

• 7,413 items, 8 nutrients per 100 g, no missing values. Flagged 1 duplicate row, 67 items with sugar > carbohydrate, and 56 calorie mismatches.

• Most nutrients are strongly right-skewed; about half of all items have no fiber or no sugar; water is bimodal (dry vs. moist foods).

• Strongest relationships: water vs. calories (Spearman −0.97), saturated vs. total fat (0.93), fat vs. calories (0.75), carbohydrate vs. fiber (0.67).

• The 1,181 category labels need to be grouped into broad food groups before cluster validation.

4. Action Items (before next week)

Task	Assignee	Deadline
Week 3 Q1: define each column and summarize the strong correlation groups	Jack Kelly	Nov 5, 2026
Week 3 Q2: choose the 3 best features to classify inventory and test them with a scatter plot	Anirudh Ramesh	Nov 5, 2026
Week 3 Q3: identify "misfit" / possibly mislabeled rows (start from data_quality_flags.csv)	Ikumi Uemura	Nov 5, 2026
Draft a mapping of the 1,181 categories into about 10 broad food groups (needed for Week 5)	Ikumi Uemura	Nov 5, 2026
Agree on a scaling method (z-score vs. min-max vs. log + z-score) for Week 4	All members	Next meeting
Commit Week 3 code and report to GitHub; update README; submit weekly sync	All members	Nov 5, 2026